BAS--102 APPLIED MATHEMATICS-I

L	T	P	Credits
3	1	0	3.5

Course Objective:

The course is aimed at developing the basic mathematical skills among the students of engineering that are imperative for effective understanding of engineering subjects. The main objective is to inculcate the knowledge of basic concepts of Calculus, Algebra, Complex Analysis and their applications for the solutions of engineering and mathematical problems. At the end of the course, students shall be able to deal with functions of several variables, matrices, system of linear equations, improper integrals and functions of complex variables. They shall learn how to implement their mathematical knowledge to solve real world problems.

Course Content

Section-A

Matrix Algebra: Special Matrices: Hermitian matrices, Skew-Hermitian matrices, Unitary matrices and Orthogonal matrices; Rank of matrices, Rank by Echelon form; Consistency of system of homogeneous and non-homogeneous equations: Gauss Elimination method, Gauss Jordan method; Eigen values and Eigen vectors of a matrix, Elementary properties of Eigen values and Eigen vectors,Eigen values of Special Matrices, Cayley Hamilton's Theorem.

Functions of several variables: Partial derivatives, Total differential, Approximation by Total Differentials, Derivatives of composite and implicit functions,Homogenous functions, Euler's theorem, Maximumand Minimum values of functions of two and three variables, Lagrange's Method of Multipliers.

Section-B

Complex Analysis: De Moivre's Theorem, Power and roots of complex numbers,Expansion of $\cos^n\theta$, $\sin^n\theta$ in terms of $\cos n\theta$ and $\sin n\theta$ and vice-versa, Analytic functions, Necessary and Sufficient condition for a function to be analytic, CR-equations, Polar form of CR-equations, Harmonic functions, Conjugate of Harmonic functions, Laplace equation.

Improper integrals: Improper integral of First and Second kind, Absolute convergence of Improper integrals, Comparison tests (without proof), Beta and Gamma functions and their properties.

RECOMENDED BOOKS:

1. R. K. Jain and S. R. K. Iyenger, Advanced Engineering Mathematics, Narosa Publishing House, New Delhi.
2. E. Kreyzic, Advanced Engineering Mathematics,Johan Wiley & Sons, (10^{th}Edition).
3. H. K. Dass, Advanced Engineering Mathematics, S. Chand and Company.
4. B. S. Grewal, Higher Engineering Mathematics, Khanna Publisher.

Scheme of Examination

- English will be the medium of instruction and examination.
- This course will carry 100 marks of which 50 marks shall be reserved for Internal Assessment and remaining 50 marks for external end semester examination.
- The duration of final written examination of this paper shall be of three hours.
- The students shall be declared passed in the paper if he/she secures minimum 40% marks in each of the Internal Assessment and External Examinations separately.

Instructions to the External Paper Setter

- The External Paper will carry 50 marks and would be of three hours. The Question paper will be divided into three Sections, namely Section-A, Section-B and Section-C. There will be FIVE questions in each of the Sections A and B, of five (05) marks each. However, a question may be segregated into subparts. Section C is compulsory and contains TEN (10) sub-parts each of two (2) marks. Candidates will be required to attempt SIX questions by selecting three Questions from each Sections A& B.
- The paper setter shall provide detailed marking instructions and solutions to numerical problems for evaluation purpose in the separate white envelopes provided for solutions.
- The paper setters should seal the internal & external envelope properly with signatures & cello tape at proper place.

CPE--101 Computer Programming

L	T	P	Credits
3	1	0	3.5

Course Objective:

Making the students understand and learn the basics of computer how to operate it, to make familiar with the part and function of computer and its types. Second part of the course is designed to provide complete knowledge of C language. Students will be able to develop logics which will help them to create programs, applications in C. Also by learning the basic programming constructs they can easily switch over to any other language in future after completing the subject, student should be able to:

- Understand the meaning and basic components of a computer system,
- Define and distinguish Hardware and Software components of computer system,
- Design an algorithmic solution for a given problem
- Write a maintainable C program for a given algorithm.
- Trace the given C program manually.
- Write C program for simple applications of real life using structures and files.

Section A

Number System: Bit, Byte, Binary, Decimal, Hexadecimal and Octal System, Conversion from one System to another.
Binary Arithmetic: Addition, Subtraction and Multiplication.
Introduction to Computer Language: Machine Language, Assembly Language, Higher Level Language, Assembler, Compiler, Interpreter.
Introduction to Operating System: Batch Systems, Multiprogramming, Time sharing Systems, Real Time Systems, Network Operating System and Distributed Operating System.

Introduction to C: Concepts of Procedure oriented programming, Character Set, Identifiers, Keywords and Data types and storage classes.
Operators and Expressions: Arithmetic, Unary, Logical, Relational, Assignment and Conditional Operator, Associatively and Precedence of Operators
Control Structures: If, while, do-while and for loop, Nested Control Structure, Switch-case, break and Continue statements

Section B

Arrays: Single Dimensional, Multidimensional Arrays and Pointers, String reading/writing
Functions: Types of Functions, Call by Value and Call by reference, Recursion, Structures. File processing: Opening and closing data files, simple writing and reading in unformatted data files.

Object Oriented Concepts: Comparison between C and C++, structure of C++ Program, Basic Input/ Output statements, introduction to Classes and Objects, creating a class and object, accessing class members (private, public), C++ Fundamentals Concepts (Definition with example) of : Encapsulation, Function Overloading, Single level Inheritance, Polymorphism and Friend Functions.

Note: This subject is common to all branches. Only basics of C++ is covered

Recommended Books:

1. E. Balagurusamy, "Programming in C", Tata McGraw Hill
2. Yashwant Kanetkar, "Let Us C", BPB
3. B. Ram, "Computer Fundamentals", Wiley
4. P.K.Sinha, "Computer Fundamentals".
5. V. Rajaraman, " Fundamentals of Computers", PHI
6. Brain W. Kernigha and Dennis M. Richie: The C Programming Language, PHI
7. Robert Lafore, "Turbo C++"
8. E. Balagurusamy, "Programming in C++", Tata McGraw Hill

CPE--151 COMPUTER PROGRAMMING LAB

L	T	P	Credits
0	0	2	1.0

Course Objective:

To familiarize the students with basic concepts of computer programming and developer tools. To present the syntax and semantics of the "C" language as well as data types offered by the Language. To allow the students to write their own programs using standard language infrastructure regardless of the hardware or software platform. After completing the subject, student should be able to:

- Design an algorithmic solution for a given problem
- Write a maintainable C program for a given algorithm.
- Trace the given C program manually.
- Write C program for simple applications of real life using structures and files.

List of Experiments

1. Experiencing DOS internal and external commands.
2. Introducing 'C' language basics such as data types, variables, constants etc.
3. Working with operators (arithmetic, logical and relational).
4. Write a program showing input and output functions.
5. Write a program to illustrate decision control structures.
6. Write program using looping control structures.
7. Write applications based on one and two dimensional arrays.
8. Working with pointers.
9. Write a program showing array and pointer relationship.
10. Illustrate functions and recursion.
11. Show the use of pointers in functions.
12. Write a program to show the use of functions with arrays.
13. Write a program based on structure and using union.
14. Use the pointer to point to structure.
15. Use the structures with functions.
16. Illustrate the file handling.
17. Write program to illustrate C++ program structure.
18. Write program to illustrate the use of classes and objects.
19. Write program to illustrate the concept of inheritance.
20. Write program to illustrate the concept of polymorphism.

HSS--101 COMMUNICATION SKILLS

L	T	P	Credits
2	1	0	2.5

Course Objective:
The objective of the course is to hone the communicative skills of the budding engineers who are expected to be globally competent. It aims at inculcating in them the skills of effective Business communication, Reading, Writing, Listening and Speaking (LSRW) skills in English. It also aims at developing the learners' grammatical competence so as to equip them with appropriate language expressions to communicate effectively in both oral and written contexts.

Section – A

Communication: Process of communication, semantic gap, Types and channels of communication. Significance of communication in a professional organisation.

Reading and Comprehension Skills: Reading purposes, gears, types and effective strategies of reading. Strategies of comprehension, analysis and interpretation: character analysis, theme, central idea, etc.

The following text is prescribed for this unit:

Prose Parables, Orient Blackswan Private Limited-2013

Selected Texts from this book are to be studied:

1. **Kabuliwallah,** Rabindranath Taagore
2. **The Eyes are Not Here,** Ruskin Bond
3. **Grief,** Anton Chekhov
4. **The Doctor's Word,** R. K. Narayan

Writing Skills: Elements of effective writing, writing styles, one word substitution.

Business Correspondence: Elements & kinds of business letters; quotations & tenders, Job application, Résumé, Agenda, memorandum, Report writing, e-mail etiquettes.

Section – B

Listening Skills: Process of listening, kinds of listening, barriers to listening, how to become an effective listener and feedback skills.

Grammar: Transformation of sentences, Active and Passive voice, Narration, correction of Sentences

Speaking Skills: Speech Mechanism, articulation of sounds, phonetic transcription, components of effective talk, group discussion, oral presentation skills, types and use of audio visual aids in presentation.

Text and Readings:

1. Lata, Pushp and Sanjay Kumar. *Communication Skills.*2015.OUP India.
2. Raman, Meenakshi, and Sangeeta Sharma. *Technical Communication :Principles and Practice.*2011.OUP India.
3. Rizvi, M. Ashraf.*Effective Technical Communication.*2005.Tata McGraw Hill Education Pvt. Ltd.
4. Lewis,Norman.*Speak Better Write Better English.* 2011.Goyal publishers.
5. Greenbaum,Sidney.*OxfordEnglish Grammar.*2005.Oxford.
6. Martin,Wren.*High School English Grammar and Composition.* 1995.S. Chand & Company Ltd., New Delhi
7. Best,Wilfred D. *The Students' Companion* . Harper collins.

Scheme of Examination:

- English will be the medium of communication.
- The course will carry 100 marks of which 50 marks shall be reserved for Internal Assessment and remaining 50 for External End-semester examination.
- The duration of final written examination of this paper shall be three hours.

Instructions to the External Paper Setter:

- The External Paper will carry 50 marks and would be of three hours duration.The Question Paper will be divided into three Sections, namely Section-A, Section-B and Section-C.There will be FIVE questions in each of the Sections A and B. Each question will be of five (05) marks. However, a question may be segregated into subparts.Section C is compulsory and contains TEN (10) sub-parts each of two (2) marks. Candidates will be required to attempt SIX questions by selecting three Questions from each Sections A & B.
- The paper setters should seal the internal & external envelope properly with signatures & cello tape at proper place.

MCE--151 ENGINEERING GRAPHICS

L	T	P	Credits
2	4	0	4.0

Course Objective:

The objective of this course is to inculcate good understanding of basic fundamentals of engineering graphics in the students. This course is aimed at to make the student understand visual science in the form of technical graphics. General instructions related to Theory of Orthographic Projection of points, lines, planes and solids as per the BIS codes prevalent to drawing practice will be introduced initially. Section of solids, intersection and development of surfaces, isometric projection and orthographic projection of simple solids/blocks will further upgrade the basic understanding and visualization of geometrical objects and to certain extent the machine parts.

At the end of the course, the students should be able to - (1) acquire knowledge of different conventions and methods of engineering drawing, (2) understand dimensioned projections, (3) learn how to create two-dimensional images of objects using first angle orthographic projection, (4) learn how to create isometric, perspective and auxiliary projections.

Section-A

Lines, Lettering, Dimensioning, Scales; Reference and Auxiliary Planes; Systems of Orthographic Projections; Projection of Points and Lines; True length of lines and their true angles of inclination with the reference planes; Projection of Planes and their true shape.
Polyhedral and Solids of Revolution; Projection of Solids in simple positions: Axis parallel to both the reference planes, parallel or perpendicular to one and inclined to the other or inclined to both the reference planes.
Section of solids: Section Planes, Sections and projection of sections on the reference planes; True shape of sections of simple solids.

Section-B

Development of lateral surfaces of simple solids such as cubes, prisms, cylinders, pyramids, cones, spheres etc. Intersection of lateral surfaces of simple solids penetrating into one another; Projection of lines/curves of intersection/interpenetration on the reference planes.
Isometric axes, lines and planes; Isometric scale; Drawing/Sketching isometric view of planes, plane figures and simple solids from orthographic projections; Conversion of pictorial view of simple solids into orthographic projections.

Recommended Books:

1. P.S. Gill, A Text Book of Engineering Drawing (Geometrical Drawing), S.K. Kataria & Sons, New Delhi.
2. Dhananjay A. Jolhe, Engineering Drawing with an Introduction to Autocad, Tata McGraw Hill, New Delhi
3. M.B. Shah and B.C. Rana, Engineering Drawing, Pearson Education Asia, New Delhi
4. Basant Agrawal and C.M. Agrawal, Engineering Drawing, Tata McGraw Hill, New Delhi
5. French and Virck, Graphic Science, McGraw Hill Publishers, New York.
6. R. K. Dhawan, A text book of Engineering Drawing, S. Chand and Co. Ltd., New Delhi
7. N. D. Bhatt and V. M. Panchal, Engineering Drawing, Charotor Publication House, Anand
8. Venugopal, Engineering Drawing and Graphics, New Age International Publishers, New Delhi.

GROUP A: FIRST SEMESTER & GROUP B: SECOND SEMESTER

BASIC ELECTRICAL ENGINEERING

Course Code: ECEB1103T / ECEB1203T

L	T	P	Credits
3	1	0	4.0

Course Objectives:
Understanding basic concepts of Electrical Science is very important as life today, is unthinkable without the use of electrical energy. This forms the base for all the engineering disciplines. Light, Industry, Agriculture, air-conditioning, broadcasting and television systems, telephone all are dependent on electrical energy.

Course Outcomes:
After the completion of the course, the student should be able to (1) Learn basic concepts of Electrical Engineering (2) Apply network laws and theorems to solve electric circuits (3) Analyze transient and steady response of DC circuits (4) Explain and Analyze the behaviour of transformer, (5) Understand the principle and characteristics of DC motor and DC generator.

Section-A

Kirchhoff's laws and their applications in solving electrical network problems, Star-delta transformation D.C. Networks: Superposition theorem. Thevenin's theorem, Norton's theorem, Maximum power transfer theorem, Step voltage response of RL and RC series circuits.

Sinusoidal Steady-State Response of Circuits: Concept of Phasor diagram, form factor and peak factor of a waveform. Series and parallel circuits, power and power factors, Resonance in circuits, Balanced 3-phase voltage, current and power relations, 3- phase power measurement.

Section-B

Single-Phase Transformers: Constructional feature, working principle of a transformer, emf equations, Transformer on no -load and its phasor diagram, Transformer on load, voltage drops and its phasor diagram, Equivalent circuit, Ideal transformer, open and short circuit tests, Calculation of efficiency, condition for maximum efficiency
Electrical Machines: Construction, Principle of working, Function of the commutator for motoring and generation action, Characteristics and applications of DC Motor.

Text Books:

1. B.L.Theraja & A.K.Theraja, A textbook of Electrical Technology, S Chand publishers.
2. Naidu, M.S. and Kamashaiah, S., Introduction to Electrical Engineering, Tata McGraw Hill (2007).
3. Del Toro, V., Electrical Engineering Fundamentals, Prentice□Hall of India Private Limited (2004).
4. Hughes, E., Smith, I.M., Hiley, J. and Brown, K., Electrical and Electronic Technology, PHI (2008).
5. Nagrath, I.J. and Kothari, D.P., Basic Electrical Engineering, Tata McGraw Hill (2002).

Reference Books:

Chakraborti, A., Basic Electrical Engineering, Tata McGraw Hill (2008).

GROUP A: FIRST SEMESTER & GROUP B: SECOND SEMESTER
WEB DEVELOPMENT
Course Code: CSEB1110T / CSEB1210T

L	T	P	Credits
3	0	0	3.0

Course Objective:
The objective of this course is to understand the fundamental concepts of the Internet and its architecture, explain different types of Internet connections and the role of Internet Service Providers (ISPs),Describe the structure and functioning of the World Wide Web (WWW), Identify components such as web browsers, web servers, URLs, HTTP, and DNS and Gain basic skills in navigating and utilizing web technologies effectively.

Course Outcomes:
After completion of the course, the student will be able to (1) Develop a dynamic webpage (2) write a well formed / valid XML document (3) write a server side application (4) know of CSS, DHTML and PHP.

SECTION-A

INTERNET AND WORLD WIDE WEB: Introduction, Internet addressing, ISP, types of Internet connections, introduction to WWW, web browsers, web servers, URL, HTTP, DNS, web applications, tools for web site creation.

HTML: Introduction to HTML, lists, adding graphics to HTML page, creating tables, linking documents, frames, DHTML and cascading style sheets.

XML: Why XML, XML syntax rules, XML elements, XML attributes, XML DTD displaying XML with CSS.

SECTION-B

PHP: Introduction, syntax, variables, statements, operators, decision making, loops, arrays, strings, Error Handling , Try, catch, throw, forms, get and post methods, functions, cookies, sessions.

PHP and MySQL: Introduction to MySQL, connecting to MySQL database, creation, insertion, deletion and retrieval of MySQL data using PHP, PHP and XML, XML parsers, XML DOM.

Recommended Books :

1. Head First HTML with CSS & XHTML by Elisabeth Freeman and Eric Freeman
2. HTML5 for Web Designers by Jeremy Keith
3. HTML5 & CSS3 For The Real World by Alexis Goldstein, Estelle Weyl, and Louis Lazaris
4. The Essential Guide to CSS and HTML Web Design by Craig Grannell
5. The Truth About HTML5 by Luke Stevens and RJ Owen
6. XML black book by Natanya Pitts-Moultis
7. XML for Dummies by Ed Tittel and Lucinda Dykes